

09USATST7

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Problem

Find all triples (x, y, z) of real numbers that satisfy the system of equations

$$\begin{cases} x^3 = 3x - 12y + 50, \\ y^3 = 12y + 3z - 2, \\ z^3 = 27z + 27x. \end{cases}$$

Solution. Let $x = 2 + a, y = 2 + b, z = 2 + c$.

We have the following set of equations:

$$a(a + 3)^2 = -12b$$

$$b(b + 6)^2 = 3c$$

$$c(c + 9)^2 = 27a$$

If $abc \neq 0$ then multiplying all of the equations and cancelling out abc results in $(a + 3)^2(b + 6)^2(c + 9)^2 = -972$ which is impossible.

If any of a, b, c is 0 then just plugging it in in their equations give the values of the other 2 variables. Checking for all of the cases, we always get $(a, b, c) = (0, 0, 0)$.

So, $(x, y, z) = (2, 4, 6)$ is the only triplet that works.